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# SkyEvo: Generative probabilistic solar forecasting with sky image evolution

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## HIGHLIGHTS

- Probabilistic solar forecasting is achieved via stochastic sky-image evolution.
- Sky trajectories are generated to form predictive distributions at various timesteps.
- The cross-site generalizability is assessed via different datasets.
- SkyEvo shows potential in both deterministic and probabilistic forecasting.

## ARTICLE INFO

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## ABSTRACT

Accurate short-term solar irradiance forecasting is essential for the reliable operation of photovoltaic (PV) systems. However, probabilistic methods often estimate output uncertainty without explicitly representing the multiple possible evolutions of future cloud fields, which limits their ability to characterize rapidly changing solar conditions. This study presents SkyEvo, a two-stage probabilistic forecasting framework that integrates stochastic sky-image generation with multi-horizon irradiance prediction. In the first stage, a Generative Sky Image Evolution (GSE) module based on variational latent dynamics generates multiple future sky-image trajectories from historical observations, thereby representing uncertainty in future visual conditions. In the second stage, each generated trajectory is combined with irradiance or PV-power measurements through a multimodal forecasting module, producing an empirical predictive distribution from 15 stochastic samples. SkyEvo is evaluated over 10–60 min forecasting horizons on two datasets: the Folsom dataset for global horizontal irradiance forecasting and the Stanford dataset for PV power forecasting. On the Folsom dataset, SkyEvo maintains PICP above 89% across all forecasting horizons, achieves a 10-min nRMSE of 14.51%, and provides stable interval widths and quantile calibration. The GSE module also achieves lower MSE and higher PSNR and UQI than VanillaVAE and a lightweight diffusion-based video forecasting baseline. Under direct Folsom-to-Stanford transfer without fine-tuning, SkyEvo achieves the highest PICP at five of the six forecasting horizons and the lowest CRPS at all horizons, while maintaining competitive prediction-interval sharpness. These results demonstrate that stochastic future sky evolution can provide forecast-relevant visual scenarios and support reliable sample-based probabilistic forecasting across different datasets and forecasting targets.

## 1. Introduction

Solar photovoltaic (PV) energy has become a central pillar of global decarbonization strategies, driven by substantial technological progress, cost reductions, and large-scale deployment over the past decades [1,2]. Nevertheless, the intrinsic stochasticity of solar irradiance continues to pose significant operational challenges for modern power systems. Rapid PV fluctuations influence balancing markets, reserve procurement, voltage regulation, and curtailment strategies. In systems with high PV penetration, these fluctuations may trigger voltage flicker or even system instability under extreme conditions [3–5]. Such risks are

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exacerbated by minute-level cloud-induced irradiance ramps, which evolve far more rapidly than variations driven by solar geometry and are therefore considerably more difficult to model and control [6]. Consequently, accurate and uncertainty-aware intra-hour solar forecasting has become essential for reliable grid operation, optimal scheduling, and risk-sensitive decision-making. In this context, a forecasting model should not only provide an accurate central estimate, but also quantify the range, sharpness, and statistical reliability of possible future solar-generation outcomes.

Various sensing modalities have been explored for short-term solar forecasting, including satellite imagery, all-sky imagers, and ground-based sensor networks. Among these, sky cameras have emerged as a particularly powerful source of information due to their high spatial resolution, frequent capture rate, and direct observation of cloud motion and occlusion processes [7,8]. These capabilities enable detailed characterization of cloud advection, deformation, dissipation, and vertical development, which govern minute-scale irradiance transitions. Despite these advantages, the majority of sky image-based forecasting studies have focused on deterministic prediction [9–12]. Deterministic forecasts provide only a single expected trajectory and fail to quantify uncertainty, limiting their applicability in operational scenarios that require probabilistic reserves, ramp-rate control, or economic dispatch [13]. As power systems increasingly emphasize reliability and risk assessment, probabilistic forecasting has become a critical research and operational requirement [14–16]. A central challenge is that sky image-based probabilistic forecasting must represent uncertainty in both future cloud-field evolution and the corresponding irradiance or PV power. However, many existing methods estimate output uncertainty directly from historical measurements or visual features without explicitly propagating multiple possible future sky trajectories.

Although probabilistic methods have made substantial progress in satellite imagery, numerical weather prediction, and time-series forecasting, sky-image-based probabilistic solar forecasting remains comparatively underdeveloped. Early studies introduced uncertainty through ensemble learning, resampling, or hybrid statistical models. For example, Chu et al. constructed irradiance prediction intervals using hybrid machine-learning models, while Pedro et al. combined irradiance measurements and sky-image-derived information to obtain probabilistic intra-hour forecasts [17,18]. These approaches provide useful uncertainty estimates, but their predictive distributions are formed from statistical learning or resampling rather than from explicit stochastic modeling of future visual conditions.

Prediction-interval and quantile-based methods provide a more direct alternative. DenLUBE estimates lower and upper irradiance bounds from sky image-related inputs [19], whereas SkyNet directly predicts several predefined conditional quantiles from historical sky images and irradiance sequences [20]. Such approaches avoid repeated ensemble construction and can provide effective uncertainty bounds. Nevertheless, interval-based methods describe uncertainty mainly through two boundaries, while multi-quantile methods represent the predictive distribution only at selected probability levels. More importantly, neither paradigm explicitly generates multiple possible future cloud-field evolutions before estimating the associated solar uncertainty.

Recent generative forecasting studies have begun to model future visual evolution more explicitly. PredRNN++-based forecasting predicts future sky images and derives deterministic GHI, DHI, and DNI estimates using the resulting visual sequences [21]. SkyGPT employs stochastic future-video generation and maps the generated scenarios to probabilistic PV power forecasts at the Stanford site [22]. ECLIPSE jointly predicts future irradiance, associated uncertainty, and segmented future sky images over short horizons [23], while Omnivision combines ground-based sky images, satellite observations, and historical irradiance for deterministic and probabilistic forecasts up to 60 min [24]. More recently, multimodal generative AI and Transformer architectures have used multiple generated sky-image sequences together with historical solar power data to support probabilistic multi-horizon forecasting [25].

These studies demonstrate the value of combining future visual information with solar forecasting, but they differ substantially in input modality, forecasting target, prediction horizon, probabilistic formulation, dataset, and evaluation protocol. Some methods emphasize deterministic future-image prediction, some require additional satellite observations, and others are evaluated only for a specific forecasting target or horizon. Table 1 summarizes these methodological differences.

As summarized in Table 1, existing approaches generally represent solar uncertainty through prediction intervals, predefined quantiles, multimodal uncertainty estimation, or stochastic future-image generation. However, interval and quantile methods do not explicitly propagate uncertainty in future visual evolution, while existing generative approaches are often limited to a single forecasting horizon, additional sensing modalities, or specific forecasting targets. A remaining need is therefore a framework that can generate multiple possible future sky trajectories and consistently propagate them into multi-horizon empirical solar predictive distributions.

To address this need, we propose SkyEvo, a two-stage generative probabilistic framework for intra-hour solar forecasting. The primary objective is to estimate multi-horizon predictive distributions by sampling possible future sky-image evolution trajectories and propagating their visual uncertainty into irradiance or PV power forecasting. In the first stage, a Generative Sky Image Evolution (GSE) module based on variational latent dynamics generates multiple plausible future sky-image sequences conditioned on historical observations. Rather than explicitly resolving or separately quantifying physical atmospheric processes, GSE learns data-driven stochastic spatiotemporal patterns of sky evolution, including changes in cloud-field appearance, spatial configuration, and solar occlusion that are relevant to short-term solar forecasting. In the second stage, each generated trajectory is combined with historical solar measurements through a ResNet18-based visual encoder, an LSTM-based temporal encoder, and a compact multilayer decoder. The forecasts produced from the multiple generated trajectories form an empirical predictive distribution at each horizon.

The main contributions of this study are summarized as follows:

1. A Generative Sky Image Evolution (GSE) module is developed to learn stochastic spatiotemporal sky dynamics and to generate multiple plausible future sky-image trajectories from identical historical observations, thereby explicitly representing uncertainty in future cloud-field evolution.
2. A two-stage multimodal probabilistic forecasting framework is proposed to combine each generated sky trajectory with recent irradiance or PV power measurements. The resulting stochastic solar trajectories are aggregated to construct empirical predictive distributions over multiple forecasting horizons.
3. The proposed framework is systematically evaluated over 10–60 min horizons on the Folsom irradiance and Stanford PV power datasets. The evaluation covers probabilistic and deterministic forecasting performance, calibration reliability, generated-image quality, generative baseline comparisons, architectural ablation, and cross-dataset generalizability.

The rest of this article is organized as follows. Section 2 introduces the proposed methodology and evaluation metrics. Section 3 presents the experimental results and corresponding discussion. Section 4 concludes the study and outlines its limitations and future research directions.

## 2. Methodology

To address the aforementioned challenges, we propose SkyEvo, a stochastic video generation model-based two-stage probabilistic solar forecasting framework. The proposed SkyEvo framework consists of two sequential stages. In the first stage, historical sky images are processed using a stochastic video generation model to predict the evolution of

**Table 1**
Comparison of representative sky-image-based deterministic and probabilistic solar forecasting methods.

| Method | Input modality | Probabilistic strategy | Horizon | Dataset | Reported metrics | Main limitation relative to the present task |
|---|---|---|---|---|---|---|
| Chu et al. [17] | Sky images and DNI measurements | Prediction intervals generated using hybrid SVM and ANN models | 5–20 min | Folsom | PI coverage and width; deterministic forecast errors | No generation of multiple future sky-image trajectories or sample-based predictive distributions. |
| Pedro et al. [18] | GHI/DNI measurements and sky-image features | kNN- and gradient-boosting-based probabilistic forecasts | Intra-hour | Folsom | RMSE, FS, PICP, PINAW, CRPS, and reliability analysis | Uncertainty estimation from measurements and image-derived features without explicit future sky-scene generation. |
| DenLUBE [19] | Ground-based sky images and irradiance-related variables | Lower-upper-bound prediction intervals | Intra-hour | Author-collected | Coverage, interval width, and deterministic errors | Prediction intervals without multiple future sky scenarios or sample-based empirical distributions. |
| SkyNet [20] | Historical sky images and irradiance sequences | Direct multi-quantile regression | 10–60 min | Folsom | PICP, PINAW, CRPS, nRMSE, FS, and RS | Conditional quantile prediction without modeling uncertainty in future sky-image evolution. |
| ECLIPSE [23] | Historical sky images and irradiance information | Joint irradiance, uncertainty, and segmented-image prediction | Up to 10 min | SIRTA | Irradiance, temporal-lag, uncertainty, and image metrics | Short-horizon segmented-image and irradiance prediction without stochastic RGB sky-trajectory sampling. |
| Omnivision [24] | Sky images, satellite observations, and past irradiance | Deterministic and probabilistic multimodal forecasting | Up to 60 min | SIRTA | Deterministic and probabilistic forecasting metrics | Dependence on additional satellite observations and absence of sampled future ground-based RGB trajectories. |
| PredRNN++ [21] | Historical sky images and sun-blocking index | None; deterministic image and irradiance forecasting | 10 min | Author-collected | nRMSE for GHI, DHI, and DNI | No probabilistic prediction intervals, predictive samples, or calibration evaluation. |
| SkyGPT [22] | Historical sky images and PV-power measurements | Stochastic video generation and sample-based PV forecasting | 15 min | Stanford | CRPS, Winkler score, and image-generation metrics | Single-horizon evaluation and separated video generation and PV forecasting stages. |
| Multimodal GenAI-Transformer [25] | Historical sky images and solar-power sequences | Generated future sky sequences with Transformer-based probabilistic forecasting | 15–60 min | Kuitun | RMSE, CRPS, and interval-based probabilistic metrics | Specific solar-power forecasting setting with limited evaluation across acquisition conditions. |
| **SkyEvo** | **Historical sky images and irradiance or PV-power sequences** | **Stochastic latent sky evolution with sample-based empirical distributions** | **10-60 min** | **Folsom and Stanford** | **PICP, PINAW, CRPS, calibration, nRMSE, FS, RS, MSE, PSNR, and UQI** | **Generation of multiple future sky trajectories and propagation of visual uncertainty into multi-horizon irradiance and PV-power forecasts.** |

future sky image sequences. In the second stage, the predicted sky image sequences, together with historical irradiance measurements, are utilized to forecast future solar irradiance through an encoder-decoder architecture. The overall framework is illustrated in Fig. 1.

Specifically, the model uses four historical RGB sky images as visual context and produces six future irradiance values corresponding to the 10–60 min forecasting horizons. The first stage transforms the historical image sequence into multiple possible future sky image trajectories, while the second stage maps each generated trajectory and the recent irradiance history into a future irradiance trajectory. Repeating this procedure for multiple latent samples yields an empirical predictive distribution at each forecast horizon.

### 2.1. Generative sky image evolution

To model the stochastic evolution of cloud fields and their visual transformations over time, a generative forecasting framework termed GSE is developed. The GSE model extends the stochastic video generation paradigm to the domain of sky cameras by integrating convolutional encoders, recurrent dynamics, and variational inference. This design enables the model to effectively capture both the spatial morphology of clouds and the temporal uncertainty inherent in atmospheric motion. A detailed illustration of GSE is shown in Fig. 2. In implementation, GSE accepts an input tensor of historical sky images with shape $(B, T_{\text{in}}, C, H, W)$, where $T_{\text{in}} = 4$, $C = 3$, and $H = W = 256$. The model predicts $T_{\text{out}} = 6$ future sky frames. The same implementation also supports the channel-first temporal format $(B, C, T_{\text{in}}, H, W)$, which is internally permuted to the time-first format before encoding.

To generate future sky evolutions, a sequence of historical sky images is first processed through a downsampling encoder consisting of $N$ convolutional layers. In this study, three convolutional layers with channel dimensions of (64, 128, 128) are employed. This encoder extracts multi-scale spatial representations that encode cloud texture, optical density, and luminance gradients, while preserving the local spatial coherence necessary for realistic sky reconstruction. The first two convolutional blocks use $4 \times 4$ kernels with stride 2 and padding 1, reducing the image resolution from $256 \times 256$ to $64 \times 64$. The third block uses a $3 \times 3$ kernel with stride 1 and padding 1 to refine the feature representation without further spatial downsampling. Each convolutional block consists of convolution, batch normalization, and ReLU activation. The resulting feature tensor has 128 channels and is used as the input to the recurrent dynamics module. Subsequently, a recurrent network is utilized to model temporal dependencies across frames, consisting of a Convolutional Long Short-Term Memory (ConvLSTM) module. Unlike conventional LSTMs, ConvLSTM performs convolutional operations within its gating mechanisms, enabling the preservation of spatial structure during temporal modeling. Through this recurrent

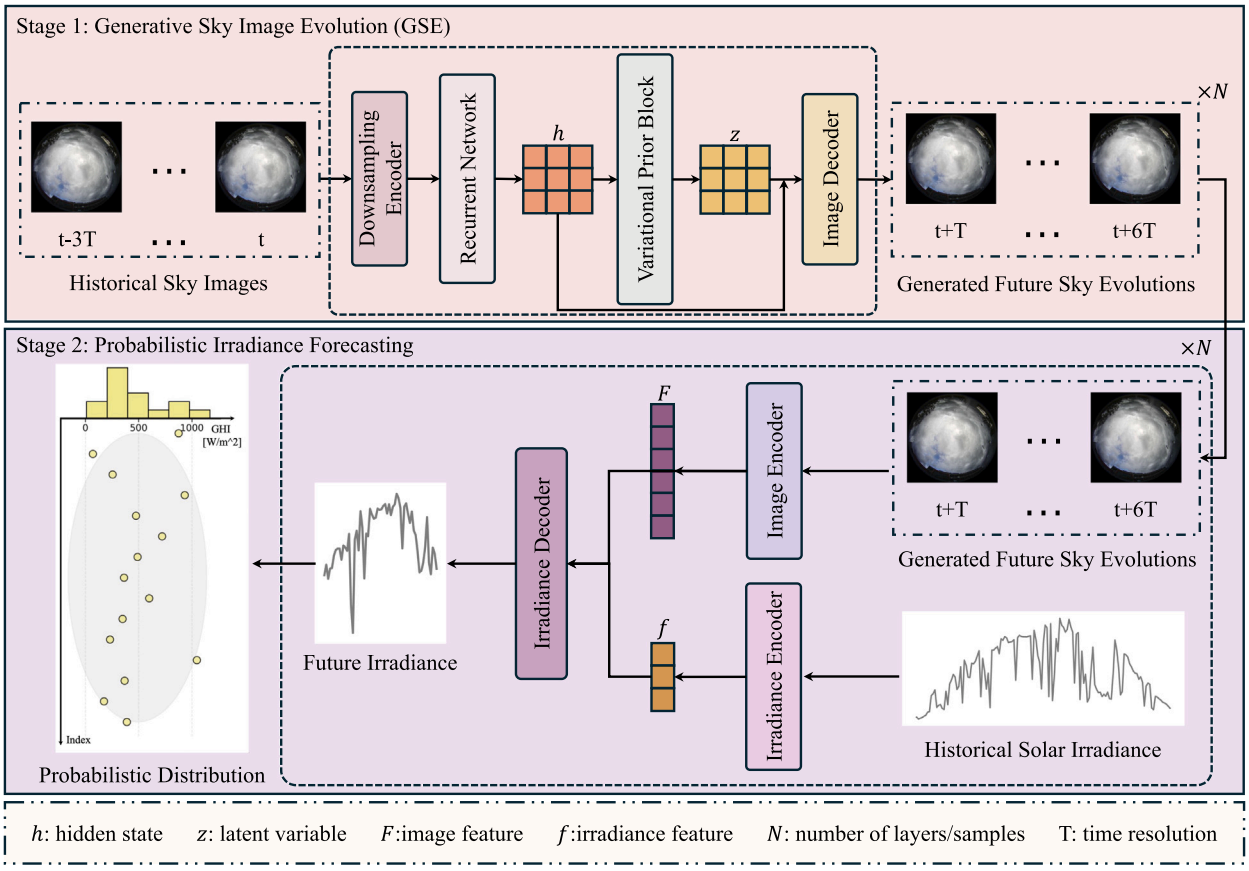



**Fig. 1.** Overall architecture of the proposed SkyEvo framework. The framework consists of two sequential stages that are jointly trained in an end-to-end manner. In stage 1, a GSE model predicts the evolution of future sky image sequences based on historical observations. For stage 2, the generated future sky images are integrated with historical solar irradiance measurements to estimate the probabilistic distributions of future irradiance.

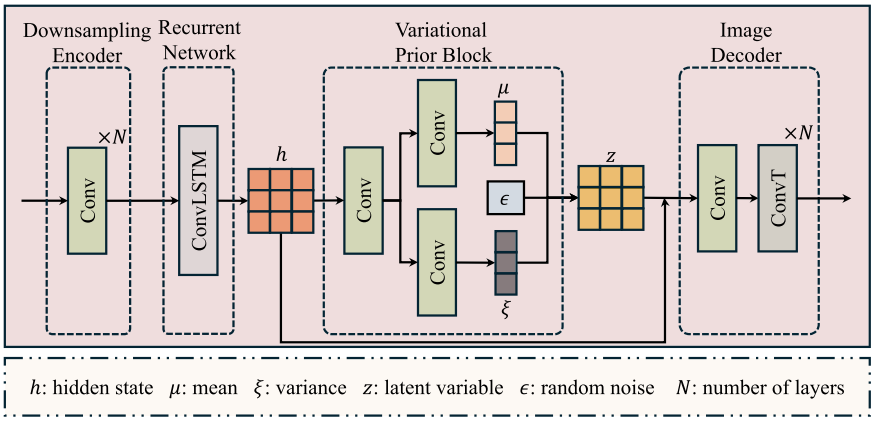



**Fig. 2.** The structure of GSE.

mechanism, the model learns temporal visual patterns associated with cloud displacement and morphological changes, allowing the hidden state to encode temporally coherent sky-image representations. In this implementation, the internal channel size of the ConvLSTM is set to 128. The ConvLSTM cell uses a $3 \times 3$ convolution with padding 1 and receives the concatenation of the encoded frame feature and the previous hidden state. The hidden state and memory cell are initialized as zero tensors with the same spatial size as the encoded feature map. The four historical frames are sequentially encoded and passed through the ConvLSTM, and the final hidden state is used as the context representation for probabilistic future generation. The basic structure of ConvLSTM is illustrated in Fig. 3. The formulas for ConvLSTM are expressed as:

$$i_t = \sigma\left(W_{xi} * x_t + W_{hi} * h_{t-1} + b_i\right), \tag{1}$$

$$f_t = \sigma\left(W_{xf} * x_t + W_{hf} * h_{t-1} + b_f\right), \tag{2}$$

$$g_t = \tanh\left(W_{xg} * x_t + W_{hg} * h_{t-1} + b_g\right), \tag{3}$$

$$o_t = \sigma\left(W_{xo} * x_t + W_{ho} * h_{t-1} + b_o\right), \tag{4}$$

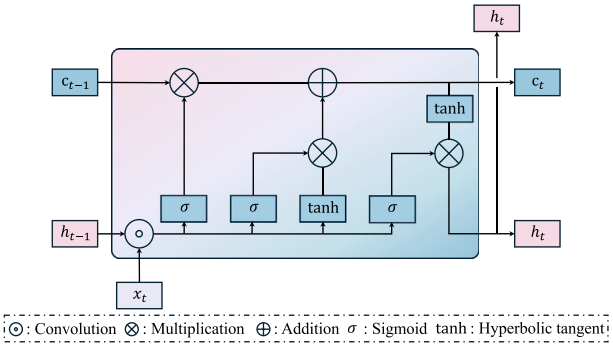



**Fig. 3.** The structure of ConvLSTM.

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t, \tag{5}$$

$$h_t = o_t \odot \tanh(c_t), \tag{6}$$

where the operator $*$ denotes the convolution operation, and $\odot$ represents the Hadamard product. Let $x_t \in \mathbb{R}^{C_x \times H \times W}$ be the input tensor at time step $t$, $h_t \in \mathbb{R}^{C_h \times H \times W}$ the hidden state, and $c_t \in \mathbb{R}^{C_c \times H \times W}$ the memory cell. The learnable convolutional kernels $W_x$ and $W_h$ correspond to input-to-state and state-to-state transitions, respectively, and $b$ denotes the associated bias terms. The functions $\sigma(\cdot)$ and $\tanh(\cdot)$ denote the logistic sigmoid and hyperbolic tangent nonlinearities.

To account for the stochastic and multimodal nature of future sky evolution, GSE incorporates a variational prior block. From the final recurrent feature map $h$, the prior block estimates the mean and log-variance maps $(\mu, \log \xi)$. Spatial averaging produces the latent distribution parameters $(\mu, \log \xi) \in \mathbb{R}^D$, where $D$ is the latent dimension. In this study, the latent dimension is set to $D = 128$. The prior block first applies a $3 \times 3$ convolution and ReLU activation to the final ConvLSTM hidden state, followed by two separate $1 \times 1$ convolutional layers that generate the spatial mean and log-variance maps. Global average pooling over the spatial dimensions converts these maps into the latent vectors $\mu$ and $\log \xi$. A latent vector $z$ is then sampled via the reparameterization trick:

$$z = \mu + \epsilon \cdot \sqrt{\xi}, \quad \epsilon \sim \mathcal{N}(0, 1). \tag{7}$$

This stochastic sampling enables the generation of multiple plausible sky evolution trajectories under identical initial conditions, thereby capturing the probabilistic characteristics of short-term atmospheric dynamics.

The latent vector $z$ is concatenated with the ConvLSTM hidden state $h$ and fed into a decoder network to synthesize future sky frames. The decoder comprises a series of transposed convolutional layers that progressively upsample the latent features into high-resolution sky images. More specifically, $z$ is spatially tiled to match the resolution of $h$ and concatenated with $h$ along the channel dimension, producing a decoder input with $128 + D$ channels. The decoder first applies a convolutional block that maps the concatenated tensor to 128 channels, followed by two transposed convolutional layers with $4 \times 4$ kernels, stride 2, and padding 1. These layers upsample the feature map back to the original image resolution. A final sigmoid activation constrains the generated RGB frame to the normalized image range. Generated frames are recursively re-encoded and passed back into the ConvLSTM, forming an autoregressive closed-loop mechanism that promotes temporal coherence across the generated frames. During autoregressive generation, each predicted frame is encoded by the same convolutional encoder and used to update the ConvLSTM hidden and memory states before generating the next future frame. This design allows the six future frames to be generated sequentially rather than independently.

During the generation stage, GSE produces multiple stochastic future sky-image trajectories by varying the scale of the latent perturbation. Given the historical sky-image sequence $x$, the encoder parameterized by $\phi$ estimates an approximate posterior distribution

$$q_\phi(z \mid x) = \mathcal{N}(\mu, \mathrm{diag}(\xi)), \tag{8}$$

where $z$ denotes the latent variable, and $\mu$ and $\xi$ denote the predicted mean deviation and variance of the latent distribution, respectively. The prior distribution is defined as $p(z) = \mathcal{N}(0, I)$, where $I$ is the identity matrix. For the $n$th stochastic sample, the latent variable is obtained as

$$z_n = \mu + \lambda_n \cdot \epsilon_n \cdot \sqrt{\xi}, \tag{9}$$

where $z_n$ denotes the $n$th sampled latent variable, $\epsilon_n$ is independently sampled Gaussian noise, and $\lambda_n$ is the latent perturbation scaling factor associated with the $n$th sample. For $N$ stochastic samples, the scaling factors are uniformly assigned within a predefined range as

$$\lambda_n = \lambda_{\min} + \frac{n-1}{N-1}(\lambda_{\max} - \lambda_{\min}), \qquad n = \{1, 2, \ldots, N\}, \tag{10}$$

where $\lambda_{\min} = 1$, $\lambda_{\max} = 3$, and $N$ denotes the number of stochastic samples. In this study, $N = 15$ stochastic samples are used, and the sensitivity analysis supporting this selection is provided in Appendix A. The scaling factor $\lambda_n$ controls the magnitude of the latent perturbation and enables the generated trajectories to represent different plausible patterns of future sky-image evolution under the same historical context. It is not a trainable parameter and is not included in the optimization objective. The resulting trajectories are subsequently passed through the irradiance forecasting module to construct an empirical predictive distribution.

### 2.2. Probabilistic irradiance forecasting

Building upon the stochastic sky sequence generation, the second stage focuses on probabilistic irradiance forecasting. The core objective is to integrate visual atmospheric evolution with temporal irradiance patterns from historical measurements. This fusion enables the model to infer irradiance trajectories that are consistent with the learned visual evolution patterns and recent solar trends. For each generated future sky trajectory, the irradiance forecasting module outputs six irradiance values corresponding to the same six forecast horizons. Therefore, for $N$ generated sky trajectories, SkyEvo obtains $N$ irradiance trajectories. The predictive distribution at each horizon is represented empirically by these $N$ samples. The number of stochastic trajectories is set to $N = 15$.

The predicted sky sequences from the GSE model encapsulate stochastic cloud morphologies that directly influence irradiance variability. To capture these high-level features, a modified ResNet-18 network is employed as the image encoder, which is shown in Fig. 4. The first convolutional layer is adapted to accept multiple concatenated frames, enabling implicit temporal feature extraction across predicted sequences. Because each generated sky trajectory contains six RGB frames, the input to the ResNet18 visual encoder is reshaped from $(B, 6, 3, 256, 256)$ to

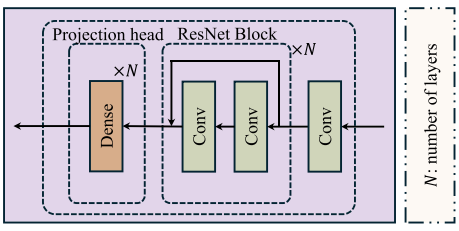



**Fig. 4.** The structure of image encoder in probabilistic irradiance forecasting stage.

($B$, 18,256,256). The original ImageNet-pretrained ResNet18 first convolution is therefore replaced by an 18-channel convolution. Its weights are initialized by repeating the pretrained three-channel kernels across the six frames and averaging them, allowing the model to reuse low-level visual filters while accepting multi-frame sky inputs. The classification head of ResNet18 is removed and replaced by a projection head with dimensions of 512, 1024, and 2000, where each projection layer is followed by batch normalization and ReLU activation. The encoded representation, $F$, summarizes the spatiotemporal evolution of the atmosphere, acting as a stochastic visual prior for irradiance estimation.

In parallel, historical irradiance observations are processed by an irradiance encoder consisting of a multilayer perceptron followed by an LSTM network. Specifically, the MLP is composed of three fully connected layers with channel sizes of 32, 128, and 256, which progressively project the raw irradiance inputs into a higher-dimensional feature space. The encoded features are then fed into a single-layer LSTM with 256 hidden units to capture the temporal dependencies within the irradiance sequence. This encoder captures both local irradiance fluctuations and longer-term periodic patterns, complementing the visual representation. The resulting feature $f$ provides a compact summary of recent irradiance dynamics that are not directly observable from sky imagery. In the implementation, the irradiance input consists of four historical irradiance values. These values are first mapped by the MLP to a 256-dimensional representation and are then processed by a one-layer LSTM with hidden size 256. The final LSTM output is used as the irradiance-history feature.

The outputs of the two encoders are concatenated to form a unified multimodal representation. This fusion captures the cross-modal dependencies between cloud evolution and irradiance variability. The fused feature is passed through a fully connected irradiance decoder with batch normalization and dropout for regularization. The irradiance decoder is implemented as a five-layer multilayer perceptron with channel sizes of 2048, 512, 128, 32, and 6, respectively, which progressively expands the latent representation into the final multi-horizon irradiance outputs. The hierarchical decoder structure enables the model to learn both global irradiance trends and fine-grained fluctuations. By introducing stochasticity during the generation of future sky evolution stages, the SkyEvo can also estimate forecast uncertainty, yielding probabilistic irradiance distributions. The fused representation has a dimension of 2256. This fused feature is subsequently processed by the irradiance decoder, which consists of fully connected layers with dimensions of 2048, 512, 128, and 32, followed by a six-dimensional output layer corresponding to the six forecasting horizons. Dropout layers with a rate of 0.2 are applied after the first, second, and fourth hidden layers for improved generalization.

Given a generated sky-image trajectory and the corresponding historical irradiance representation, the irradiance decoder deterministically produces a future irradiance trajectory. Therefore, the predictive uncertainty is introduced through stochastic sky-trajectory generation. By applying the decoder to multiple stochastic sky-image trajectories, multiple irradiance realizations are obtained and aggregated to form an empirical predictive distribution for quantile estimation and uncertainty evaluation.

### 2.3. Loss function

The SkyEvo framework is optimized using a unified loss function that jointly supervises both the generative sky image evolution and the probabilistic irradiance forecasting stages.

Although the framework contains two sequential processing stages, the two stages are not trained separately. During each training iteration, the GSE module first generates stochastic future sky-image sequences from the historical sky-image context. The generated sequences are then encoded and integrated with the historical irradiance measurements by the irradiance forecasting module to produce the corresponding multi-horizon irradiance forecasts. All trainable components of the GSE and irradiance forecasting modules are jointly updated through end-to-end backpropagation.

For image-level supervision, the GSE module is trained to reconstruct future sky images while regularizing the learned latent distribution. The corresponding training objective combines the mean squared error (MSE) reconstruction loss and the Kullback-Leibler (KL) divergence and is defined as

$$\mathcal{L}_{\mathrm{GSE}} = \mathrm{MSE}(\hat{X}_t, X_t) + \beta D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p(z)), \tag{11}$$

where $X_t$ and $\hat{X}_t$ denote the ground-truth and generated future sky images at time $t$, respectively. The first term measures the image reconstruction error, whereas the second term regularizes the approximate posterior distribution $q_\phi(z \mid x)$ toward the standard Gaussian prior $p(z) = \mathcal{N}(0, I)$. The coefficient $\beta$ controls the contribution of the KL-divergence term and is set to 0.001 based on the sensitivity analysis presented in Appendix B.

For the diagonal Gaussian posterior $q_\phi(z \mid x) = \mathcal{N}(\mu, \mathrm{diag}(\xi))$, the KL-divergence term is calculated as

$$D_{\mathrm{KL}}(q_\phi(z \mid x) \,\|\, p(z)) = \frac{1}{2} \sum_{j=1}^{d} (\mu_j^2 + \xi_j - 1 - \log(\xi_j)), \tag{12}$$

where $j$ indexes the latent dimensions, $d$ denotes the dimensionality of the latent space, and $\mu_j$ and $\xi_j$ denote the mean deviation and variance of the $j$th latent dimension, respectively.

For irradiance-level supervision, the probabilistic irradiance forecasting module is trained using an MSE loss to minimize the discrepancy between the predicted and observed irradiance values:

$$\mathcal{L}_{\mathrm{irr}} = \mathrm{MSE}(\hat{y}_t, y_t), \tag{13}$$

where $\hat{y}_t$ and $y_t$ represent the predicted and ground-truth irradiance at time $t$, respectively. The MSE loss is computed over the six forecast horizons. During inference, the trained irradiance decoder is evaluated repeatedly using the $N$ stochastic sky trajectories produced by GSE, and the resulting $N$ irradiance outputs are aggregated into empirical prediction intervals and quantiles.

To ensure end-to-end optimization, both stages are trained jointly through a unified objective function defined as:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{GSE}} + \mathcal{L}_{\mathrm{irr}}. \tag{14}$$

During backpropagation, the gradients associated with the irradiance forecasting loss are propagated through the irradiance forecasting module and further back into the GSE module. Therefore, the GSE module and the irradiance forecasting module are optimized jointly rather than being trained separately or sequentially.

This integrated loss encourages the generation of visually consistent sky image evolutions that not only align with the observed cloud dynamics but also enhance the accuracy of downstream irradiance forecasting. In practice, the two losses correspond to complementary supervision signals: image-level supervision constrains the plausibility and temporal coherence of generated sky evolution, whereas irradiance-level supervision aligns the generated visual trajectories with the downstream forecasting objective.

### 2.4. Error metrics

To assess probabilistic forecasting performance, three complementary metrics are utilized: the prediction interval coverage probability (PICP), the prediction interval normalized average width (PINAW), and the continuous ranked probability score (CRPS) [6]. For the calculation of PICP and PINAW, the prediction interval is defined by the 5th and 95th percentiles of the predictive distribution, corresponding to a nominal coverage level of 90%. The PICP measures the proportion of observed

values that fall within the predicted intervals and therefore serves as a fundamental indicator of forecast reliability. It is defined as

$$\mathrm{PICP} = \frac{1}{n'}\sum_{t=1}^{n'} \zeta_t \times 100\%, \tag{15}$$

where $\zeta_t$ equals 1 when the observation at time $t$ lies within the corresponding lower and upper bounds of the prediction interval, and equals 0 otherwise.

Complementing the coverage measure, the PINAW quantifies the average width of the prediction intervals after normalization by the data range. This metric evaluates the magnitude of predictive uncertainty and is expressed as

$$\mathrm{PINAW} = \frac{1}{n'}\frac{\sum_{t=1}^{n'}(U_t - L_t)}{G_{max} - G_{min}} \times 100\%, \tag{16}$$

where $G_{max}$ and $G_{min}$ denote the maximum and minimum observed values, respectively, and $U_t$ and $L_t$ are the upper and lower limits of the prediction interval.

As an integrated measure of probabilistic accuracy, the CRPS evaluates the discrepancy between the cumulative distribution functions of the forecasts and the observations. It is defined as

$$\mathrm{CRPS} = \frac{1}{n'}\sum_{t=1}^{n'}\int_0^{\infty}\left(F^{\hat{G}_t}(x) - \mathbb{1}(x - G_t)\right)^2 dx, \tag{17}$$

where $\mathbb{1}(x - G_t)$ represents the Heaviside step function shifted to the observed value $G_t$, and $F^{\hat{G}_t}$ denotes the predicted cumulative distribution function. Lower CRPS values indicate higher overall probabilistic forecasting precision, thus enabling a unified assessment of probabilistic and time-series forecasting performance.

To evaluate the performance of models on deterministic forecasting, three metrics are employed: the normalized root-mean-square error (nRMSE), the forecasting skill (FS), and the ramp score (RS) [12]. The nRMSE is a widely used accuracy indicator that emphasizes larger forecasting errors and is therefore more sensitive to outlier behavior. It is defined as

$$\mathrm{nRMSE} = \frac{\sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\hat{G}_i - G_i\right)^2}}{\frac{1}{n}\sum_{i=1}^{n} G_i} \times 100\%, \tag{18}$$

where $G_i$ and $\hat{G}_i$ denote the measured and predicted clear-sky indices at time $i$.

The FS quantifies the improvement of a forecasting model relative to a reference model and is expressed as

$$\mathrm{FS} = \left(1 - \frac{\mathrm{nRMSE}_{\mathrm{model}}}{\mathrm{nRMSE}_{\mathrm{per}}}\right) \times 100\%, \tag{19}$$

where $\mathrm{nRMSE}_{\mathrm{per}}$ and $\mathrm{nRMSE}_{\mathrm{model}}$ represent the nRMSE values of the persistence model and the forecasting model, respectively. In this study, the clear-sky persistence model serves as the reference.

The RS is formulated as

$$\mathrm{RS} = \frac{1}{t_{\max} - t_{\min}}\int_{t_{\min}}^{t_{\max}} \left|\mathrm{SD}(T(t)) - \mathrm{SD}(R(t))\right| dt, \tag{20}$$

where SD denotes the swinging door compression output, which characterizes ramp slopes. The terms $t_{\max}$ and $t_{\min}$ specify the period during which the global horizontal irradiance remains positive. $T(t)$ and $R(t)$ correspond to the forecasted and measured time-series data.

Within these evaluation criteria, lower nRMSE and RS values indicate improved forecasting performance and an enhanced capability in identifying critical ramp events. In contrast, higher FS values reflect superior model performance relative to the baseline.

To assess the quality of the generated future sky images, three widely used image-evaluation metrics are adopted: Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and the Universal Quality Index (UQI) [26]. These metrics provide complementary evaluations of reconstruction performance from the perspectives of pixel-wise error, signal fidelity, and structural similarity.

MSE measures the average squared difference between the generated image and its corresponding ground-truth image, thereby providing a direct indication of pixel-level reconstruction error. A lower MSE value represents a closer agreement between the predicted and reference images. Based on MSE, PSNR expresses the reconstruction quality on a logarithmic scale by comparing the maximum possible image intensity with the residual reconstruction error. For 8-bit sky images, PSNR is calculated as

$$\mathrm{PSNR}(\hat{I}, I) = 10\log_{10}\left(\frac{255^2}{\mathrm{MSE}(\hat{I}, I)}\right), \tag{21}$$

where $I$ and $\hat{I}$ denote the ground-truth and generated images, respectively. A higher PSNR generally indicates better preservation of the image's dynamic range and lower reconstruction distortion.

In addition to pixel-level fidelity, UQI evaluates the perceptual similarity between two images by jointly considering luminance consistency, contrast similarity, and structural correlation. Its value reaches one when the generated image is identical to the reference image and decreases as discrepancies in brightness, contrast, or spatial structure increase. This characteristic makes UQI particularly suitable for evaluating whether cloud shapes and spatial patterns are preserved in generated ground-based sky images. UQI is defined as

$$\mathrm{UQI} = \frac{4\delta_{xy}\bar{x}\bar{y}}{(\delta^2 * x + \delta^2 * y)[(\bar{x})^2 + (\bar{y})^2]}, \tag{22}$$

where $\bar{x}$ and $\bar{y}$ represent the mean intensities of the reference and generated images, $\delta_x^2$ and $\delta_y^2$ denote their corresponding variances, and $\delta_{xy}$ is their covariance. These quantities are computed as

$$\begin{cases} \bar{x} = \dfrac{1}{N}\sum_{i=1}^{N} I_i, \\ \bar{y} = \dfrac{1}{N}\sum_{i=1}^{N} \hat{I}_i, \\ \delta_x^2 = \dfrac{1}{N-1}\sum_{i=1}^{N}(I_i - \bar{x})^2, \\ \delta_y^2 = \dfrac{1}{N-1}\sum_{i=1}^{N}(\hat{I}_i - \bar{y})^2, \\ \delta_{xy} = \dfrac{1}{N-1}\sum_{i=1}^{N}(I_i - \bar{x})(\hat{I}_i - \bar{y}). \end{cases} \tag{23}$$

## 3. Results and discussion

### 3.1. Folsom dataset and experimental settings

The dataset utilized in this study was acquired in Folsom, California, USA (38.642°, 121.148°, GMT), using a rooftop-mounted fisheye sky camera in conjunction with a rotating shadowband radiometer, as detailed in [27]. It comprises ground-based sky images (GSIs) paired with corresponding global horizontal irradiance (GHI) measurements, totaling approximately 500,000 samples captured at one-minute intervals between 2014 and 2016. The dataset covers a wide range of clear-sky, partly cloudy, and overcast conditions, thereby providing substantial variability in cloud appearance, solar occlusion, and irradiance fluctuation patterns. It has been widely adopted as a benchmark for intra-hour

sky image-based irradiance forecasting and is used in this study to evaluate the capability of SkyEvo to model stochastic short-term sky evolution and its associated irradiance uncertainty. To account for seasonal variability, data collected from January 2014 to July 2015 are allocated for training, those from August to December 2015 for validation, and the entire 2016 dataset for testing. This temporal partitioning ensures a clear chronological separation between model training and evaluation phases.

In the forecasting configuration, the model receives as input a temporal sequence of four consecutive sky images acquired at 10-minute intervals ($\mathcal{R} = 10$ minutes), with each frame resized to a spatial resolution of 256 × 256 pixels. To comprehensively evaluate forecasting capability, SkyEvo is assessed under six prediction horizons $\mathcal{H} \in \{10, 20, 30, 40, 50, 60\}$ minutes, representing critical timescales for solar energy forecasting and grid management.

All experiments are implemented in the PyTorch deep learning framework and executed on a computing server equipped with 8 NVIDIA RTX 3090 GPUs. The model is trained over 60 epochs, with the test dataset kept completely separate from both training and validation procedures. Optimization is carried out using the Adam optimizer, while a Reduce-on-Plateau learning rate scheduler dynamically adjusts the learning rate. The initial learning rate is set to 0.01 and is reduced by a factor of 0.5 whenever the validation loss fails to improve for five consecutive epochs, thereby promoting stable and efficient convergence.

### 3.2. Forecasting performance on the folsom dataset

#### 3.2.1. Probabilistic forecasting performance

Building on this setup, the probabilistic forecasting results in Table 2 show that SkyEvo achieves consistently strong performance across the complete 10–60 min forecasting range. Its PICP remains above 89% at all forecasting horizons and stays above 90% within the first 40 min, decreasing only moderately to approximately 89.3% at the 60-min horizon. This indicates that the proposed method maintains reliable interval coverage as forecasting uncertainty increases. Among the evaluated methods, SkyEvo achieves the highest PICP at the 20-, 30-, 40-, 50-, and 60-min horizons, whereas $\text{SUNSET}_q$ obtains a marginally higher PICP at 10 min, with values of 92.82% and 92.60%, respectively. SkyGPT also achieves competitive coverage at the shorter horizons, reaching 92.07% and 91.10% at 10 and 20 min, but its PICP decreases more substantially with the forecasting horizon, falling to 79.80% at 50 min and 72.55% at 60 min. In comparison, $\text{SUNSET}_e$ exhibits substantially lower coverage, while $\text{ViT}_q$ and $\text{CNNLSTM}_q$ show more pronounced horizon-dependent variations. Although $\text{SUNSET}_q$ achieves competitive coverage at selected horizons, this is accompanied by substantially wider prediction intervals, as reflected by its consistently large PINAW values.

**Table 2**
Performance of models for probabilistic forecasting.

| Metric | Model | $\mathcal{H} = 10$ | $\mathcal{H} = 20$ | $\mathcal{H} = 30$ | $\mathcal{H} = 40$ | $\mathcal{H} = 50$ | $\mathcal{H} = 60$ |
|---|---|---|---|---|---|---|---|
| PICP | $\text{SUNSET}_e$ | 77.21 | 71.06 | 63.91 | 66.85 | 58.40 | 65.11 |
| | $\text{SUNSET}_q$ | **92.82** | 91.22 | 90.06 | 89.50 | 88.08 | 86.62 |
| | $\text{CNNLSTM}_q$ | 86.80 | 87.91 | 87.38 | 86.97 | 86.41 | 85.24 |
| | $\text{ViT}_q$ | 82.16 | 83.58 | 86.81 | 84.28 | 81.58 | 78.51 |
| | SkyGPT | 92.07 | 91.10 | 89.72 | 85.27 | 79.80 | 72.55 |
| | **SkyEvo** | 92.60 | **91.39** | **90.94** | **90.30** | **89.54** | **89.26** |
| PINAW | $\text{SUNSET}_e$ | 23.11 | **21.01** | 20.58 | 20.69 | **19.79** | **20.17** |
| | $\text{SUNSET}_q$ | 41.29 | 37.92 | 37.65 | 37.70 | 38.81 | 39.18 |
| | $\text{CNNLSTM}_q$ | **22.98** | 21.33 | **20.26** | **20.07** | 20.84 | 23.22 |
| | $\text{ViT}_q$ | 29.99 | 26.00 | 27.17 | 28.75 | 31.49 | 34.85 |
| | SkyGPT | 25.06 | 26.29 | 25.20 | 27.14 | 25.72 | 26.86 |
| | **SkyEvo** | 24.74 | 23.50 | 23.77 | 24.12 | 23.85 | 24.25 |
| CRPS | $\text{SUNSET}_e$ | 75.09 | 74.95 | 75.51 | 75.64 | 77.05 | **76.92** |
| | $\text{SUNSET}_q$ | 119.04 | 117.45 | 117.30 | 118.23 | 124.69 | 126.14 |
| | $\text{CNNLSTM}_q$ | **73.12** | **72.25** | **69.80** | **70.23** | **74.36** | 81.38 |
| | $\text{ViT}_q$ | 93.04 | 87.87 | 90.42 | 96.45 | 107.11 | 117.95 |
| | SkyGPT | 74.19 | 75.90 | 76.88 | 74.53 | 77.22 | 78.47 |
| | **SkyEvo** | 73.89 | 75.42 | 76.43 | 78.10 | 79.21 | 80.49 |

SkyEvo, by contrast, maintains comparatively narrow and stable prediction intervals across all forecasting horizons. $\text{CNNLSTM}_q$ and $\text{SUNSET}_e$ produce relatively narrow intervals at several short horizons, but this improved sharpness is achieved at the expense of insufficient empirical coverage. SkyGPT produces wider prediction intervals than SkyEvo at all six forecasting horizons, while simultaneously exhibiting lower PICP values. For example, at the 60-min horizon, SkyEvo achieves a PICP of 89.26% with a PINAW of 24.25%, whereas SkyGPT obtains a substantially lower PICP of 72.55% despite producing a wider PINAW of 26.86%. This comparison indicates that SkyEvo provides a more favorable combination of interval coverage and sharpness than SkyGPT.

The CRPS results reveal a different ranking from that observed for PICP and PINAW. $\text{CNNLSTM}_q$ achieves the lowest CRPS from 10 to 50 min, whereas $\text{SUNSET}_e$ obtains the lowest value at 60 min. SkyEvo therefore does not achieve the minimum CRPS at any individual forecasting horizon. Nevertheless, its CRPS increases gradually from 73.89 at 10 min to 80.49 at 60 min, without the pronounced long-horizon deterioration observed for $\text{SUNSET}_q$ and $\text{ViT}_q$. Compared with SkyGPT, SkyEvo obtains lower CRPS at the 10-, 20-, and 30-min horizons, whereas SkyGPT achieves lower values at 40, 50, and 60 min. These results indicate that the primary advantage of SkyEvo lies in its stable interval coverage and sharpness rather than uniform superiority in distributional accuracy.

To further assess the statistical reliability of the sampling-based predictive distributions, Fig. 5 presents quantile calibration curves across all forecasting horizons. For each nominal quantile level, the empirical frequency is computed as the proportion of ground-truth observations that fall below the corresponding predicted quantile. A perfectly calibrated predictive distribution would yield empirical frequencies equal to their nominal probabilities and would therefore coincide with the diagonal reference line. Across the evaluated horizons, the calibration curves of SkyEvo generally remain close to the ideal diagonal, indicating a consistent correspondence between the predicted quantile levels and the observed frequencies. This result provides evidence that the sampled irradiance trajectories constitute a statistically meaningful approximation of the conditional predictive distribution. Moderate calibration deviations are nevertheless observed at several quantile levels, particularly at longer forecasting horizons, where the increasing uncertainty and complexity of future sky evolution make accurate distributional estimation more challenging.

The comparison methods exhibit comparatively less consistent calibration patterns. $\text{SUNSET}_e$ and $\text{SUNSET}_q$ show pronounced departures from the diagonal at several lower and intermediate quantile levels, suggesting systematic discrepancies between their nominal probabilities and empirical occurrence frequencies. $\text{CNNLSTM}_q$ and $\text{ViT}_q$ achieve favorable calibration at selected quantile levels but exhibit greater horizon-dependent variability. By comparison, SkyEvo maintains a more stable calibration pattern across both the evaluated quantile range and forecasting horizons. This analysis complements the PICP, PINAW, and CRPS results by evaluating reliability over the complete predictive distribution rather than through a single prediction interval. Taken together, the results indicate that the favorable coverage achieved by SkyEvo is not attributable solely to interval widening, but is accompanied by comparatively consistent agreement between nominal quantile probabilities and their empirical frequencies.

These quantitative behaviors are further summarized by the radar visualization in Fig. 6. With max–min normalization highlighting the relative differences among the evaluated metrics, SkyEvo occupies a more favorable region of the metric space by jointly achieving high empirical coverage, moderate interval width, and competitive CRPS. $\text{SUNSET}_e$ produces relatively narrow but poorly calibrated intervals, whereas $\text{SUNSET}_q$ trades interval sharpness for increased coverage. $\text{ViT}_q$ and $\text{CNNLSTM}_q$ provide more balanced profiles than the SUNSET variants

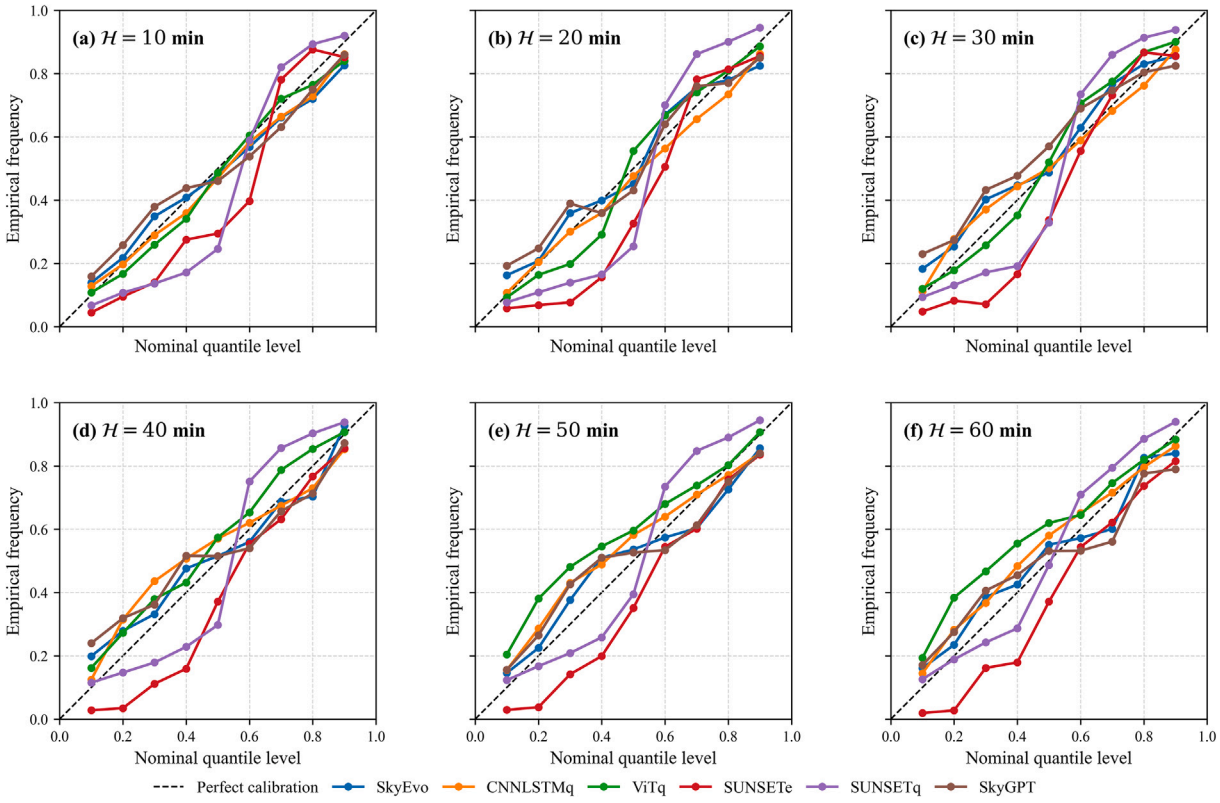



**Fig. 5.** Calibration curves of SkyEvo and the comparison methods on the Folsom dataset over forecasting horizons from 10 to 60 min. The horizontal axis denotes the nominal quantile level, while the vertical axis represents the corresponding empirical frequency of observations falling below the predicted quantile. The dashed diagonal line indicates perfect calibration, and curves closer to the diagonal represent better agreement between the predicted probabilities and observed frequencies.

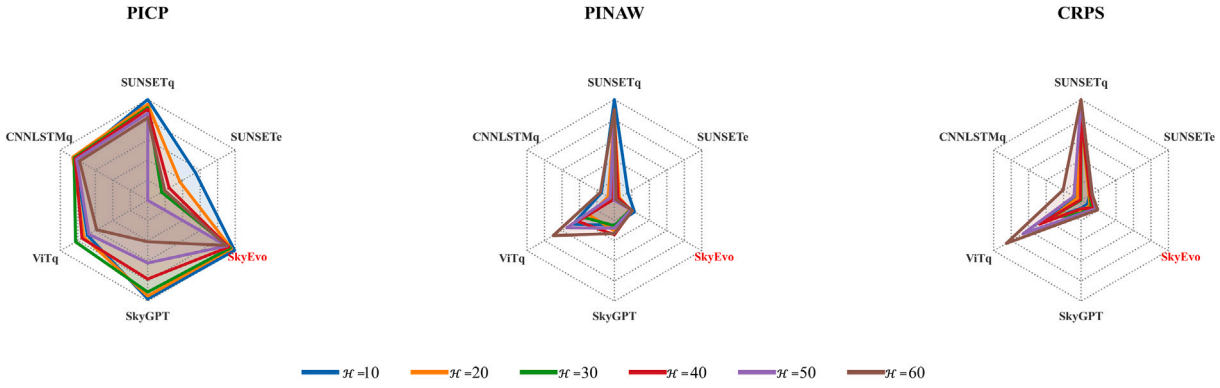



**Fig. 6.** Radar chart comparison of probabilistic forecasting performance across different models and prediction horizons. Each axis represents one forecasting model, and each polygon corresponds to a specific horizon. The metrics include PICP, PINAW, and CRPS, where higher PICP and lower PINAW or CRPS indicate better performance.

but still exhibit weaker stability across forecasting horizons. SkyGPT achieves competitive coverage and distributional accuracy at the shorter forecasting horizons; however, its coverage deteriorates substantially as the forecasting horizon increases, and its prediction intervals remain wider than those of SkyEvo across all six horizons.

The principal advantage of SkyEvo is therefore not an isolated optimum in every individual metric, but a more stable balance among interval reliability, sharpness, calibration, and distributional forecasting accuracy across the complete 10–60 min forecasting range. In particular, its relatively stable PINAW values indicate that the maintained empirical coverage is not achieved by continuously expanding the prediction intervals as the forecasting horizon increases. Although $\mathrm{CNNLSTM_q}$ and $\mathrm{SUNSET_e}$ achieve lower CRPS values at several horizons, and SkyGPT remains competitive in terms of CRPS, none of these methods simultaneously maintain the same level of empirical coverage and interval-width stability as SkyEvo over the complete forecasting range.

Fig. 7 illustrates the stochastic future sky-image trajectories generated by GSE under cloudy and clear-sky conditions. Under cloudy conditions, the samples represent multiple possible developments of the observed cloud field, including translational displacement, changes in cloud morphology, merging or separation, and gradual dissipation. Under clear-sky conditions, the generated trajectories remain comparatively stable and avoid introducing substantial artificial cloud structures. These qualitative observations indicate that the stochastic generator

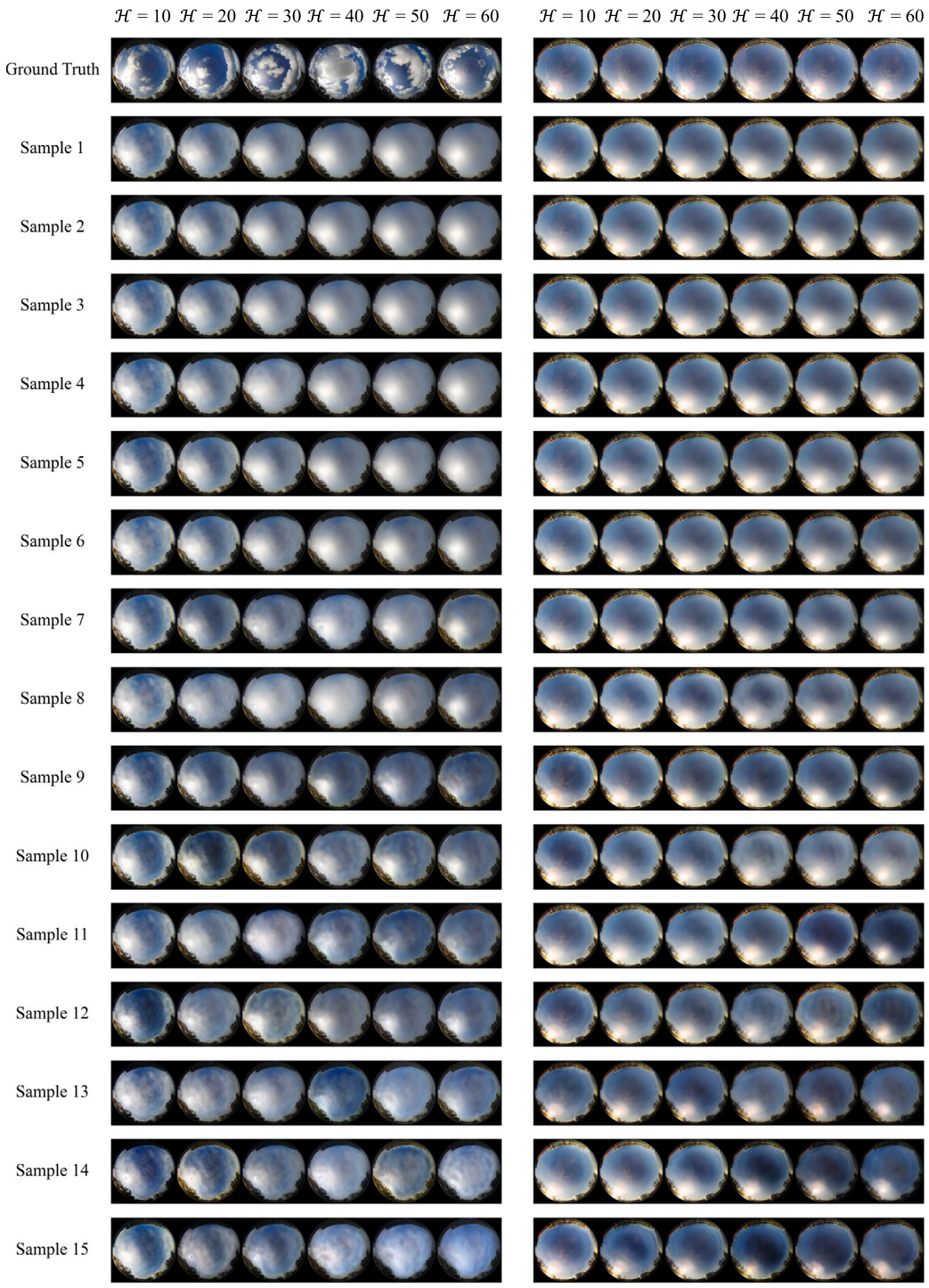



**Fig. 7.** Generated future sky-image trajectories under cloudy and clear-sky conditions. Fifteen stochastic trajectories are shown across the six forecasting horizons.

adapts the diversity of its outputs to the variability of the observed sky conditions.

To visually corroborate these aggregate results, Fig. 8 compares probabilistic prediction curves for three cloudy days and one sunny day. Under cloudy conditions, the baseline models frequently exhibit temporal lag, excessive smoothing, or insufficient interval coverage around rapid irradiance ramps. SkyEvo captures a larger proportion of these short-term variations while adapting the prediction interval width to the evolving cloud conditions. Its intervals expand around rapidly changing irradiance periods and remain comparatively narrow during more stable periods, demonstrating that the predicted uncertainty responds to the underlying cloud dynamics.

Under sunny conditions, where the future irradiance trajectory is comparatively deterministic, all methods perform more similarly. Nevertheless, SkyEvo maintains narrow and stable uncertainty bounds without introducing unnecessary spread. These case studies further demonstrate that the stochastic generator does not impose a uniform level of uncertainty across all weather conditions, but instead produces adaptive predictive distributions according to the temporal variability of the observed sky conditions.

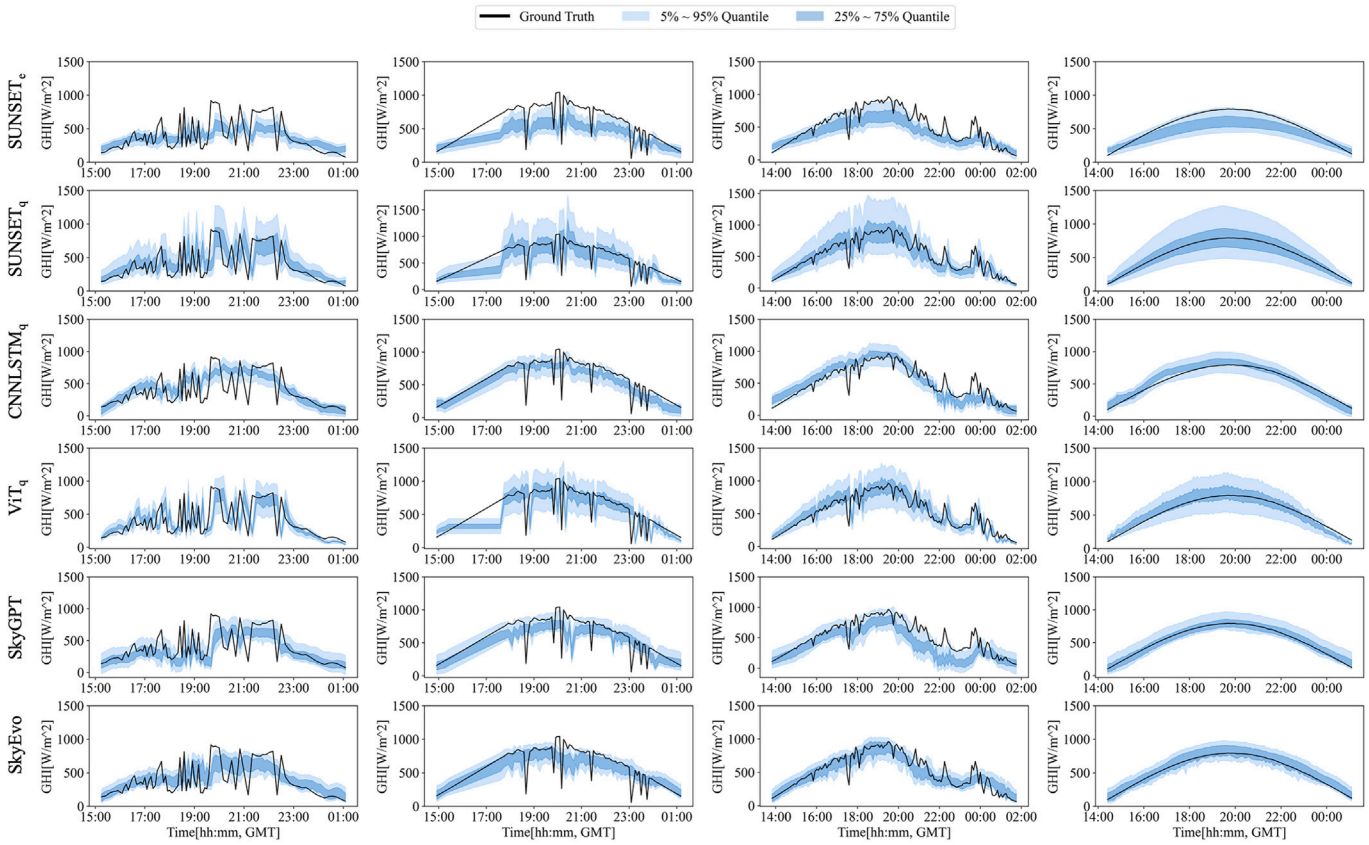



**Fig. 8.** Prediction curves for three cloudy days and one sunny day in the test dataset for different forecasting methods on 10 minute-ahead probabilistic solar irradiance forecasting.

### 3.2.2. Deterministic forecasting performance

While the above discussion emphasizes probabilistic performance, Table 3 provides a horizon-dependent comparison of the deterministic forecasting results, using the median forecast where applicable. SkyEvo achieves the lowest nRMSE from 10 to 50 min, with values increasing from 14.51% at 10 min to 18.99% at 50 min. At the 60-min horizon, CNNLSTM$_q$ obtains a slightly lower nRMSE than SkyEvo, with values of 20.07% and 20.21%, respectively. Compared with SkyGPT, SkyEvo achieves lower nRMSE values across all forecasting horizons.

**Table 3**
Performance of models on deterministic forecasting.

| Metric | Model | $\mathcal{H}=10$ | $\mathcal{H}=20$ | $\mathcal{H}=30$ | $\mathcal{H}=40$ | $\mathcal{H}=50$ | $\mathcal{H}=60$ |
|---|---|---|---|---|---|---|---|
| nRMSE | SUNSET$_e$ | 16.68 | 18.33 | 20.22 | 20.99 | 22.73 | 22.73 |
| | SUNSET$_q$ | 16.77 | 18.51 | 19.15 | 19.29 | 20.90 | 21.94 |
| | CNNLSTM$_q$ | 16.06 | 16.84 | 17.49 | 18.54 | 19.28 | **20.07** |
| | ViT$_q$ | 16.53 | 17.34 | 17.97 | 19.03 | 19.98 | 21.05 |
| | SkyGPT | 17.90 | 18.09 | 18.50 | 20.83 | 20.75 | 21.34 |
| | **SkyEvo** | **14.51** | **15.54** | **16.44** | **17.71** | **18.99** | 20.21 |
| RS | SUNSET$_e$ | 23.70 | 26.78 | 31.67 | 31.58 | 34.97 | 33.44 |
| | SUNSET$_q$ | 23.81 | 28.08 | 29.24 | 25.73 | 29.34 | 31.07 |
| | CNNLSTM$_q$ | 20.38 | 21.18 | 21.92 | **23.24** | **24.04** | **24.68** |
| | ViT$_q$ | 21.61 | 22.41 | 23.04 | 24.12 | 25.24 | 26.64 |
| | SkyGPT | 28.84 | 28.21 | 27.87 | 32.92 | 31.30 | 31.04 |
| | **SkyEvo** | **18.08** | **19.67** | **21.54** | 23.38 | 25.92 | 28.28 |
| FS | SUNSET$_e$ | 2.91 | 12.02 | 15.74 | 25.99 | 30.36 | 38.58 |
| | SUNSET$_q$ | 2.38 | 11.18 | 20.20 | 31.98 | 35.96 | 40.70 |
| | CNNLSTM$_q$ | 6.51 | 19.16 | 27.12 | 34.63 | 40.92 | **45.77** |
| | ViT$_q$ | 3.77 | 16.77 | 25.09 | 32.90 | 38.77 | 43.11 |
| | SkyGPT | −4.79 | 12.84 | 22.56 | 26.27 | 36.22 | 42.14 |
| | **SkyEvo** | **15.04** | **25.15** | **31.17** | **37.32** | **41.64** | 45.20 |

For the ramp score, SkyEvo achieves the lowest values at 10, 20, and 30 min, whereas CNNLSTM$_q$ obtains the lowest values at 40, 50, and 60 min. A similar horizon-dependent pattern is observed for FS: SkyEvo achieves the highest FS from 10 to 50 min, while CNNLSTM$_q$ obtains the highest value at 60 min. SkyEvo also consistently outperforms SkyGPT in both RS and FS across all six forecasting horizons. These results show that the relative performance of SkyEvo and CNNLSTM$_q$ varies across forecasting horizons and evaluation metrics, while SkyEvo maintains consistently better deterministic performance than SkyGPT.

These deterministic results complement the probabilistic evaluation. The nRMSE and FS results obtained by SkyEvo from 10 to 50 min indicate that the median of its empirical predictive distribution provides an accurate central estimate over these horizons. However, the results do not indicate uniform superiority over all competing models across every metric and forecasting horizon.

### 3.2.3. Generated sky evolution and generative baseline comparison

To provide a broader comparison across alternative generative paradigms, GSE is compared with a VanillaVAE and a diffusion-based video forecasting model, denoted as Diffusion-VF. VanillaVAE generates future sky images by sampling from a learned latent distribution but does not explicitly model recurrent temporal dependencies across the forecast sequence. Diffusion-VF instead conditions an iterative denoising process on the historical sky-image observations and generates future sky-image sequences from Gaussian noise. For a controlled comparison, all three generative models use the same historical inputs, forecasting horizons, image resolution, stochastic sample number, downstream forecasting architecture, data partition, and evaluation protocol.

The generated trajectories are further evaluated quantitatively against the corresponding future observations using MSE, PSNR, and

**Table 4**
Future sky-image generation performance of the evaluated generative models on the Folsom dataset.

| Model | MSE | PSNR | UQI |
|---|---|---|---|
| **GSE** | **264.29 ± 287.68** | **25.52 ± 3.45** | **0.96 ± 0.02** |
| VanillaVAE | 472.95 ± 328.25 | 22.04 ± 2.18 | 0.79 ± 0.03 |
| Diffusion-VF | 639.32 ± 384.17 | 20.09 ± 1.77 | 0.71 ± 0.06 |

**Table 5**
Probabilistic forecasting performance obtained using different future sky-image generation models on the Folsom dataset.

| Metric | Model | $\mathcal{H} = 10$ | $\mathcal{H} = 20$ | $\mathcal{H} = 30$ | $\mathcal{H} = 40$ | $\mathcal{H} = 50$ | $\mathcal{H} = 60$ |
|---|---|---|---|---|---|---|---|
| PICP | **SkyEvo** | **92.60** | **91.39** | **90.94** | **90.30** | **89.54** | **89.26** |
| | VanillaVAE | 87.60 | 88.10 | 88.20 | 84.90 | 84.60 | 87.90 |
| | Diffusion-VF | 91.90 | 91.30 | 89.50 | 88.20 | 84.40 | 79.30 |
| PINAW | **SkyEvo** | **24.74** | **23.50** | **23.77** | **24.12** | **23.85** | **24.25** |
| | VanillaVAE | 30.71 | 31.58 | 32.21 | 32.94 | 33.57 | 34.08 |
| | Diffusion-VF | 26.35 | 27.14 | 28.02 | 29.17 | 30.43 | 31.61 |
| CRPS | **SkyEvo** | **73.89** | **75.42** | **76.43** | **78.10** | **79.21** | **80.49** |
| | VanillaVAE | 86.24 | 88.17 | 89.95 | 91.72 | 93.46 | 95.08 |
| | Diffusion-VF | 78.31 | 79.86 | 81.42 | 83.17 | 85.26 | 87.54 |

UQI, as reported in Table 4. GSE achieves the lowest MSE of 264.29 ± 287.68, the highest PSNR of 25.52 ± 3.45 dB, and the highest UQI of 0.96 ± 0.02. VanillaVAE produces larger pixel-level reconstruction errors and lower structural similarity, while Diffusion-VF obtains the weakest image-generation results under the adopted lightweight implementation. These results indicate that GSE more effectively preserves the luminance distribution and spatial structure of the observed future sky scenes.

The influence of the generative model on downstream probabilistic forecasting is evaluated in Table 5. When GSE is replaced by VanillaVAE, empirical coverage decreases while both prediction-interval width and CRPS increase across the forecasting horizons. This indicates that conventional latent-variable generation provides a less effective representation of forecast-relevant temporal sky evolution.

Diffusion-VF remains competitive at the shorter forecasting horizons, achieving PICP values of 91.90% and 91.30% at 10 and 20 min, respectively. However, its empirical coverage decreases progressively as the forecasting horizon increases and reaches 79.30% at 60 min. Its PINAW and CRPS values also increase more rapidly than those of SkyEvo, indicating weaker long-horizon reliability and distributional accuracy under the adopted lightweight diffusion implementation.

Taken together, the image-generation and probabilistic forecasting results show a consistent performance trend. The superior reconstruction and structural fidelity of GSE are accompanied by more reliable coverage, narrower prediction intervals, and lower distributional error in the downstream irradiance forecasts. This relationship suggests that preserving forecast-relevant visual information in the generated sky trajectories is important for constructing effective sample-based predictive distributions. Nevertheless, MSE, PSNR, and UQI evaluate image fidelity rather than the physical correctness of individual cloud processes. The results should therefore be interpreted as evidence of visually coherent and forecast-relevant sky evolution rather than as direct physical validation of cloud advection, deformation, dissipation, or occlusion.

Despite employing a generative architecture, SkyEvo remains computationally efficient, with approximately 22 million parameters, which is substantially fewer than the more than 33 million parameters required by $\text{SUNSET}_e$ and $\text{SUNSET}_q$, and considerably lower than the 93 million parameters of the ViT-based model. While $\text{CNNLSTM}_q$ has a smaller model size of 16.8 million parameters, it generally provides lower empirical coverage and a less favorable balance between interval reliability and sharpness than SkyEvo. The efficiency of SkyEvo is primarily enabled by the spatially downsampled representations used in the generative stage and the lightweight ResNet18 visual encoder. Consequently, its per-epoch runtime remains comparable to those of the evaluated deterministic and probabilistic baselines despite the inclusion of stochastic future sky-image generation.

Overall, the probabilistic, deterministic, and image-generation results demonstrate that SkyEvo provides a stable balance among distributional accuracy, central-estimate accuracy, ramp responsiveness, and generated-image fidelity. Its two-stage architecture links stochastic visual evolution with downstream irradiance forecasting, allowing uncertainty in possible future sky conditions to be propagated into multi-horizon empirical predictive distributions.

### 3.3. Ablation study of the visual and irradiance encoders

Having compared GSE with alternative generative forecasting paradigms, this subsection further examines the architectural choices within the downstream multimodal forecasting module. The complete GSE + ResNet18 + LSTM configuration is used as the reference. To evaluate the visual encoder, ResNet18 is replaced by ResNet34 or ResNet50 while all other components remain unchanged. To evaluate the irradiance-history encoder, LSTM is replaced by GRU or RNN. Each variant therefore modifies one principal encoder component while retaining the same GSE module, training data, stochastic sample number, and evaluation protocol. The results of all ablation experiments are summarized in Table 6.

The visual-encoder variants show that increasing the backbone depth does not automatically improve probabilistic forecasting. ResNet34 provides marginally higher PICP at several intermediate horizons, including 92.00% at 20 min and 90.90% at 40 min. However, these coverage values are accompanied by PINAW values of approximately 39%, which are around 15 percentage points wider than those produced by the complete SkyEvo configuration. Therefore, the marginal coverage improvement is obtained at a substantial loss of interval sharpness.

ResNet50 achieves slightly lower CRPS than ResNet18 at several longer horizons. For example, its CRPS reaches 79.30 at 60 min compared with 80.49 for the complete model. Nevertheless, its PICP decreases from 91.30% at 10 min to 77.30% at 60 min, indicating substantial long-horizon under-coverage. The lower CRPS at selected horizons should therefore not be interpreted independently of its considerably reduced empirical coverage and wider prediction intervals.

Replacing LSTM with GRU or RNN similarly weakens long-horizon reliability. The PICP of the GRU configuration decreases to 74.00% at 60 min, while that of the RNN configuration decreases to 73.00%. Both alternatives also produce substantially wider prediction intervals than the complete model. Although the recurrent alternatives remain competitive in CRPS at selected horizons, their decreasing coverage indicates that the resulting predictive distributions do not retain sufficient dispersion as uncertainty accumulates over longer forecasting horizons.

Overall, GSE + ResNet18 + LSTM provides the most consistent balance among empirical coverage, prediction-interval sharpness, and distributional accuracy. The results indicate that a deeper visual encoder does not necessarily provide more effective forecast-relevant features, while the LSTM-based irradiance encoder offers more stable long-horizon uncertainty representation than the GRU and RNN alternatives.

### 3.4. Cross-dataset transfer evaluation on the Stanford dataset

To assess the transferability of the proposed framework beyond the source domain, SkyEvo is further evaluated on the Stanford dataset [28]. The Stanford dataset was developed for short-term solar forecasting and contains quality-controlled GSIs temporally aligned with PV power measurements. This evaluation differs from the primary Folsom experiment in several respects, including geographical location, image-acquisition system, environmental conditions, and forecasting target. In particular, the forecasting target changes from GHI in Folsom to PV power output in Stanford. The Stanford experiment therefore represents both a

**Table 6**
Ablation study of the visual and irradiance-history encoders on the Folsom dataset.

| Metric | Variant | Configuration | $\mathcal{H}=10$ | 20 | 30 | 40 | 50 | 60 |
|---|---|---|---|---|---|---|---|---|
| PICP | **SkyEvo** | GSE + ResNet18 + LSTM | **92.60** | 91.39 | 90.94 | 90.30 | **89.54** | **89.26** |
| | SkyEvo-R34 | GSE + ResNet34 + LSTM | 92.20 | **92.00** | **91.30** | **90.90** | 88.40 | 83.60 |
| | SkyEvo-R50 | GSE + ResNet50 + LSTM | 91.30 | 90.30 | 89.90 | 88.40 | 83.30 | 77.30 |
| | SkyEvo-GRU | GSE + ResNet18 + GRU | 91.20 | 89.20 | 88.60 | 85.00 | 82.10 | 74.00 |
| | SkyEvo-RNN | GSE + ResNet18 + RNN | 89.60 | 88.80 | 87.70 | 85.70 | 80.60 | 73.00 |
| PINAW | **SkyEvo** | GSE + ResNet18 + LSTM | **24.74** | **23.50** | **23.77** | **24.12** | **23.85** | **24.25** |
| | SkyEvo-R34 | GSE + ResNet34 + LSTM | 39.68 | 39.68 | 39.59 | 39.62 | 39.45 | 39.18 |
| | SkyEvo-R50 | GSE + ResNet50 + LSTM | 36.30 | 36.41 | 36.64 | 36.52 | 36.25 | 35.91 |
| | SkyEvo-GRU | GSE + ResNet18 + GRU | 36.03 | 36.42 | 36.94 | 36.89 | 37.11 | 36.25 |
| | SkyEvo-RNN | GSE + ResNet18 + RNN | 37.58 | 37.25 | 37.37 | 37.35 | 37.30 | 36.66 |
| CRPS | **SkyEvo** | GSE + ResNet18 + LSTM | **73.89** | **75.42** | 76.43 | 78.10 | 79.21 | 80.49 |
| | SkyEvo-R34 | GSE + ResNet34 + LSTM | 82.74 | 82.36 | 81.32 | 81.20 | 81.85 | 83.31 |
| | SkyEvo-R50 | GSE + ResNet50 + LSTM | 75.68 | 75.76 | **75.78** | **76.01** | **77.21** | **79.30** |
| | SkyEvo-GRU | GSE + ResNet18 + GRU | 75.36 | 76.53 | 77.45 | 79.13 | 80.86 | 82.51 |
| | SkyEvo-RNN | GSE + ResNet18 + RNN | 78.32 | 77.67 | 77.45 | 77.95 | 79.92 | 82.13 |

cross-dataset and cross-target transfer setting, involving distribution shifts in the visual observations, measured variables, and their underlying relationships.

In this study, the quality-controlled samples collected in September and October 2019 are used exclusively as an unseen target-domain evaluation set, resulting in a total of 8423 valid samples. No Stanford sample is used for model training, validation, checkpoint selection, hyperparameter optimization, or fine-tuning. All model parameters and data-preprocessing statistics are determined exclusively from the Folsom training and validation subsets before evaluation on Stanford.

The forecasting configuration is kept consistent with the Folsom experiment wherever applicable. Historical sky images and the corresponding recent measurements are used as multimodal inputs, and probabilistic PV power forecasts are generated for six forecasting horizons, $\mathcal{H} \in \{10, 20, 30, 40, 50, 60\}$ min. During Folsom training, the GHI targets are max-min normalized using the minimum and maximum values of the Folsom dataset. For Stanford evaluation, the PV power measurements are normalized using the minimum and maximum values of the Stanford dataset, and the normalized model outputs are inverse-transformed using the same values to recover PV power forecasts in the original scale. SkyEvo, SkyGPT, $\text{SUNSET}_e$, $\text{SUNSET}_q$, $\text{CNNLSTM}_q$, and $\text{ViT}_q$ are trained and selected exclusively on the Folsom dataset and subsequently evaluated on the Stanford samples with their parameters fixed. This common protocol ensures that the comparison reflects the relative transferability of the different forecasting approaches under the same target-domain constraints.

The probabilistic PV power forecasting performance under the Folsom-to-Stanford transfer protocol is reported in Table 7. All models are trained and selected exclusively using the Folsom dataset and are directly evaluated on the Stanford dataset without fine-tuning, or parameter updating. Under this challenging transfer setting, SkyEvo achieves the highest PICP at the 10-, 30-, 40-, 50-, and 60-min forecasting horizons, while SkyGPT obtains the highest PICP only at the 20-min horizon. Averaged across the six horizons, SkyEvo achieves a PICP of 69.24%, compared with 68.43% for SkyGPT. These results indicate that SkyEvo preserves empirical uncertainty coverage more consistently under the substantial differences in geographical location, imaging conditions, and forecasting-target distributions between the Folsom and Stanford datasets.

The PINAW and CRPS results further demonstrate that the higher coverage achieved by SkyEvo is not obtained by indiscriminately widening the prediction intervals. SkyEvo achieves an average PINAW of 35.38%, which is 2.74 percentage points lower than the 38.12% obtained by SkyGPT. Although $\text{SUNSET}_e$ produces the smallest PINAW at five forecasting horizons, it also exhibits the lowest PICP at every horizon. Its relatively narrow intervals therefore correspond to substantial undercoverage rather than a more favorable balance between coverage and sharpness. More importantly, SkyEvo obtains the lowest CRPS at all six forecasting horizons, with an average value of 2.89, compared with 3.13 for $\text{SUNSET}_e$ and 3.82 for SkyGPT. Since CRPS evaluates the complete predictive distribution, the consistently lower values achieved by SkyEvo indicate that its probabilistic forecasts are more closely aligned with the observed Stanford targets. At the 20-min horizon, SkyGPT achieves a slightly higher PICP than SkyEvo, but it also produces a wider prediction interval and a higher CRPS. Therefore, this isolated coverage improvement does not represent an overall advantage in probabilistic forecasting quality.

**Table 7**
Probabilistic PV power forecasting performance on the Stanford dataset under Folsom-to-Stanford transfer without fine-tuning.

| Metric | Model | $\mathcal{H}=10$ | 20 | 30 | 40 | 50 | 60 |
|---|---|---|---|---|---|---|---|
| PICP | SkyEvo | **73.55** | 67.73 | **69.37** | **68.72** | **69.57** | **66.47** |
| | SkyGPT | 71.84 | **69.80** | 69.33 | 66.79 | 67.89 | 64.91 |
| | $\text{SUNSET}_e$ | 67.16 | 61.04 | 62.41 | 62.00 | 62.72 | 60.17 |
| | $\text{SUNSET}_q$ | 69.94 | 63.53 | 65.06 | 64.87 | 65.80 | 62.82 |
| | $\text{CNNLSTM}_q$ | 69.06 | 62.96 | 64.32 | 63.94 | 65.02 | 61.99 |
| | $\text{ViT}_q$ | 71.04 | 64.83 | 66.27 | 65.87 | 67.00 | 63.96 |
| PINAW | SkyEvo | 37.38 | 34.85 | 33.75 | **34.58** | 35.90 | 35.82 |
| | SkyGPT | 37.45 | 37.65 | 38.49 | 38.78 | 39.82 | 36.50 |
| | $\text{SUNSET}_e$ | **35.68** | **34.15** | **33.05** | 35.88 | **33.20** | **33.12** |
| | $\text{SUNSET}_q$ | 36.76 | 35.23 | 34.13 | 36.96 | 34.28 | 34.20 |
| | $\text{CNNLSTM}_q$ | 36.45 | 34.92 | 33.82 | 36.65 | 33.98 | 33.89 |
| | $\text{ViT}_q$ | 37.22 | 35.69 | 34.59 | 37.42 | 34.75 | 34.66 |
| CRPS | SkyEvo | **2.74** | **3.02** | **2.73** | **3.07** | **2.83** | **2.93** |
| | SkyGPT | 3.11 | 3.31 | 3.75 | 3.86 | 4.24 | 4.63 |
| | $\text{SUNSET}_e$ | 2.97 | 3.36 | 3.19 | 3.21 | 2.98 | 3.09 |
| | $\text{SUNSET}_q$ | 3.15 | 3.22 | 3.31 | 3.45 | 3.23 | 3.31 |
| | $\text{CNNLSTM}_q$ | 3.18 | 3.25 | 3.17 | 3.25 | 3.32 | 3.21 |
| | $\text{ViT}_q$ | 3.03 | 3.19 | 3.36 | 3.27 | 3.28 | 3.16 |

The performance difference may be partly attributed to the forecasting-oriented multimodal architecture of SkyEvo. Specifically, SkyEvo employs a ResNet18 encoder to extract spatial representations from historical sky images and an LSTM encoder to model the temporal evolution of historical irradiance. The two information sources are subsequently integrated so that the stochastic future sky-image trajectories are explicitly associated with the corresponding irradiance dynamics. This design enables the generated uncertainty to reflect visual changes that are relevant to the forecasting target, rather than image-level diversity alone. By comparison, SkyGPT places greater emphasis on generative modeling of future sky-image sequences. Its generated visual variations may not be coupled with historical irradiance dynamics as explicitly as in the forecasting-oriented fusion structure of SkyEvo.

Consequently, some stochastic variations in the generated images may not be effectively translated into appropriately distributed irradiance trajectories when the model is transferred to a dataset with different visual and target characteristics. This architectural distinction provides a plausible explanation for the wider intervals and substantially higher CRPS values produced by SkyGPT.

Overall, the three probabilistic metrics consistently demonstrate the stronger transfer performance of SkyEvo. It achieves the highest PICP at five of the six forecasting horizons, produces narrower prediction intervals than SkyGPT on average, and obtains the lowest CRPS at every horizon. These findings suggest that combining stochastic sky-image evolution with explicit image-irradiance feature fusion improves both uncertainty coverage and distributional accuracy under cross-dataset transfer. Nevertheless, the PICP values of all models remain below the nominal coverage level, indicating that substantial target-domain shifts still impair probabilistic calibration. Further improvements may therefore require more domain-robust distribution learning or target-agnostic calibration strategies that do not rely on Stanford labels, model fine-tuning, or target-domain parameter adaptation.

## 4. Conclusion

This study introduces SkyEvo, a two-stage probabilistic solar forecasting framework that integrates stochastic sky image generation with multi-horizon solar prediction. Rather than explicitly resolving individual atmospheric processes, SkyEvo learns data-driven stochastic spatiotemporal patterns of cloud-field evolution and solar occlusion from historical sky observations. The generated future sky-image trajectories are subsequently propagated through a multimodal forecasting module to construct empirical predictive distributions for irradiance and PV power forecasting.

On the Folsom dataset, the probabilistic forecasts maintain PICP above 89% across all horizons up to 60 min, with PINAW values of approximately 23–25%. The deterministic results exhibit a horizon-dependent pattern: SkyEvo achieves the lowest nRMSE from 10 to 50 min and the lowest RS from 10 to 30 min, whereas $\text{CNNLSTM}_{\text{q}}$ obtains a slightly lower nRMSE at 60 min and lower RS values from 40 to 60 min. The additional experiments on the Stanford dataset further demonstrate that, under direct Folsom-to-Stanford transfer without fine-tuning, SkyEvo achieves the highest PICP at five of the six forecasting horizons and the lowest CRPS at all horizons, while maintaining competitive prediction-interval sharpness.

These results highlight the ability of SkyEvo to generate empirically reliable probabilistic distributions, represent short-term irradiance variability, and maintain robust performance across multiple forecast horizons, providing a practical and effective tool for grid operators. By combining generative sky-image evolution with multimodal irradiance decoding, SkyEvo advances intra-hour solar forecasting, offering reliable, high-resolution predictions that enhance grid stability and facilitate risk-aware management of high-penetration photovoltaic systems.

Several limitations should nevertheless be acknowledged. First, although the evaluation includes two real-world datasets with different geographical locations, sensing configurations, dataset characteristics, and forecasting targets, broader validation across additional climatic conditions, camera systems, and PV installations is still required to establish wider generalizability. Second, the current framework constructs empirical predictive distributions through stochastic sampling while being trained with MSE- and KL-based objectives; therefore, it does not directly optimize a proper probabilistic scoring rule. Future work will investigate quantile-, CRPS-, and likelihood-based objectives together with post-hoc calibration methods. Finally, generating multiple future sky-image trajectories introduces additional inference cost relative to a single deterministic forward pass. Adaptive sampling, model compression, and more computationally efficient generative architectures will be explored to improve deployment efficiency.

## CRediT authorship contribution statement

**Guoping Ruan:** Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Yiheng Li:** Visualization, Software, Data curation. **Xiaoyang Chen:** Writing – review & editing, Methodology, Funding acquisition, Data curation, Conceptualization. **Maizhen Ning:** Visualization, Software. **Qi Su:** Visualization. **Fangyu Wu:** Writing – review & editing, Validation. **Yang Du:** Writing – review & editing. **Lurui Fang:** Writing – review & editing, Funding acquisition. **Eng Gee Lim:** Supervision, Resources, Project administration, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Sensitivity analysis on the selection of the stochastic sample number

In SkyEvo, the stochastic sample number determines the number of generated trajectories used to construct the empirical predictive distribution. Therefore, it directly affects the representation of predictive uncertainty and the resulting probabilistic forecasting performance.

A sensitivity analysis was conducted using five different stochastic sample numbers, including 5, 10, 15, 20, and 25 samples. For each configuration, the same trained model was used to generate stochastic future sky-image trajectories and corresponding irradiance forecasts. The latent scaling range was fixed as $[1, 3]$ for all configurations, with the scaling factors uniformly allocated within this interval. Therefore, changing the sample number modifies both the number of generated trajectories and the discrete latent scaling levels used to construct the empirical predictive distribution. The resulting ensemble forecasts were evaluated using PICP, PINAW, and CRPS, while all other model configurations and evaluation settings remained unchanged.

The results are summarized in Table A.1. The probabilistic forecasting performance does not consistently improve as the stochastic sample number increases. In particular, the 15-sample configuration produces a pronounced reduction in PINAW while maintaining relatively high PICP and achieving the lowest CRPS values. This result is related to the joint effects of the stochastic trajectory number, the discretization of the latent scaling factors, and the empirical construction of the prediction intervals. Because the interval bounds are calculated from the empirical percentiles of a relatively small number of generated trajectories, changing the sample number modifies both the discrete latent scaling levels and the empirical percentile positions used to determine the interval boundaries.

Among the evaluated configurations, the 15-sample setting produces substantially narrower prediction intervals while retaining relatively high coverage, resulting in the lowest PINAW and CRPS values and the highest or near-highest PICP values across most forecasting horizons. This result suggests that the empirical predictive distribution generated under the 15-sample configuration is more concentrated around and better aligned with the observed irradiance values. Therefore, the simultaneous reduction in PINAW and improvement in PICP are not contradictory, because narrower intervals can still achieve higher coverage when they are more accurately centered around the observations.

**Table A.1**
Sensitivity analysis of different numbers of stochastic samples in terms of PICP, PINAW, and CRPS across multiple forecasting horizons.

| Metric | Samples | $\mathcal{H} = 10$ | $\mathcal{H} = 20$ | $\mathcal{H} = 30$ | $\mathcal{H} = 40$ | $\mathcal{H} = 50$ | $\mathcal{H} = 60$ |
|---|---|---|---|---|---|---|---|
| PICP | 5 | 89.90 | 89.40 | 87.10 | 86.80 | 85.60 | 82.60 |
| | 10 | 90.60 | 90.50 | 88.70 | 90.40 | 87.20 | 85.80 |
| | **15** | **92.60** | **91.39** | **90.94** | 90.30 | **89.54** | **89.26** |
| | 20 | 91.30 | 90.80 | 89.50 | 89.70 | 87.50 | 86.80 |
| | 25 | 91.80 | 90.90 | 89.10 | **90.60** | 88.40 | 87.90 |
| PINAW | 5 | 30.85 | 31.36 | 31.80 | 32.29 | 32.52 | 32.64 |
| | 10 | 31.86 | 32.40 | 32.88 | 33.41 | 33.68 | 33.81 |
| | **15** | **24.74** | **23.50** | **23.77** | **24.12** | **23.85** | **24.25** |
| | 20 | 32.39 | 32.97 | 33.47 | 34.06 | 34.36 | 34.53 |
| | 25 | 32.59 | 33.17 | 33.68 | 34.26 | 34.56 | 34.73 |
| CRPS | 5 | 76.46 | 77.92 | 78.86 | 79.61 | 81.24 | 82.52 |
| | 10 | 78.18 | 79.33 | 80.33 | 81.11 | 82.39 | 83.84 |
| | **15** | **73.89** | **75.42** | **76.43** | **78.10** | **79.21** | **80.49** |
| | 20 | 78.81 | 80.32 | 81.28 | 82.27 | 83.48 | 84.70 |
| | 25 | 78.96 | 80.64 | 81.51 | 82.54 | 83.85 | 84.89 |

The pronounced improvement obtained with 15 samples should be interpreted as an empirical result under the adopted latent scaling and percentile-based interval construction strategy, rather than as a theoretically guaranteed advantage of using exactly 15 samples. Increasing the sample number to 20 or 25 introduces additional inference cost without providing consistent improvements in PICP, PINAW, or CRPS. Therefore, based on the observed probabilistic performance and inference cost, 15 stochastic samples were retained as the final configuration of SkyEvo.

## Appendix B. Sensitivity analysis on the selection of the KL-divergence weighting coefficient $\beta$

In SkyEvo, the KL-divergence weighting coefficient $\beta$ controls the trade-off between reconstruction accuracy and latent distribution regularization during training. Therefore, selecting an appropriate $\beta$ value is essential for maintaining both high-quality sky image-generation and effective latent representation learning.

A sensitivity analysis was conducted using four candidate values of $\beta$, including 0.1, 0.01, 0.001, and 0.0001. All models were trained under identical dataset partitions, network architectures, and optimization configurations. The generated future sky images were evaluated against the corresponding ground-truth images using MSE, PSNR, and UQI.

The results are reported in Table B.1. Among all evaluated configurations, $\beta = 0.001$ achieves the best overall image-generation performance, with the lowest MSE and the highest PSNR and UQI values. Larger $\beta$ values introduce stronger latent regularization, which may reduce reconstruction fidelity, while an excessively small $\beta$ value weakens the regularization effect and leads to degraded generation quality. Based on this analysis, $\beta = 0.001$ was selected as the final configuration of SkyEvo.

**Table B.1**
Sensitivity analysis of the KL divergence weight $\beta$ on future sky-image generation performance.

| $\beta$ | MSE | PSNR | UQI |
|---|---|---|---|
| 0.1 | 273.66 ± 290.33 | 25.29 ± 3.36 | 0.95 ± 0.06 |
| 0.01 | 268.20 ± 285.08 | 25.37 ± 3.35 | 0.95 ± 0.03 |
| 0.001 | **264.29 ± 287.68** | **25.52 ± 3.45** | **0.96 ± 0.02** |
| 0.0001 | 277.71 ± 291.74 | 25.17 ± 3.29 | 0.95 ± 0.08 |

## Data availability

The data that has been used is confidential.

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